-Lab 2 assigned today

Measuring/Computing Performance
1.4, 1.8, 1.9

Performance

- 1.4, 1.8, 1.9

- Accurately and fairly compare the performance of different systems

- Learn how to use performance metrics

Performance metrics

Computer System_X

- Execution time and Performance

$$\text{Execution Time}_X = ET_X = \frac{1}{\text{Performance}_X}$$

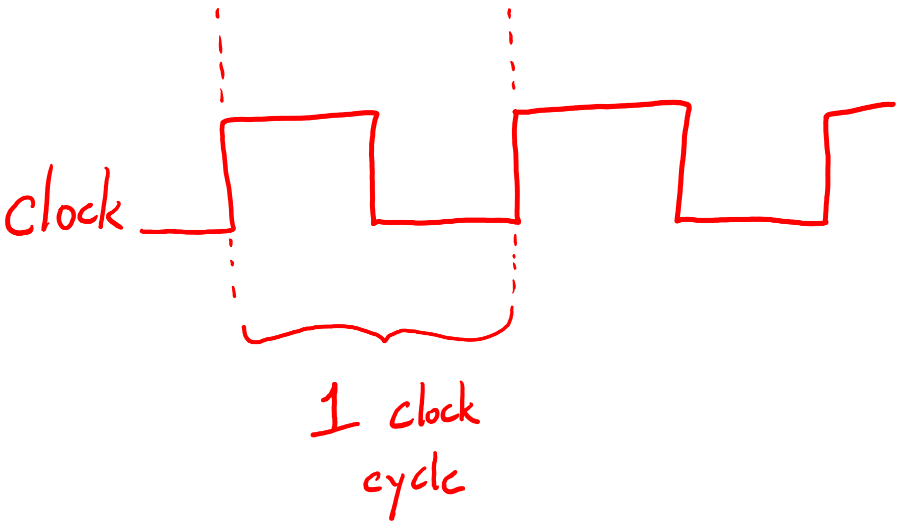
$$\text{Perf}_X = \frac{1}{ET_X}$$

- X in N times faster than Y

$$N = \frac{\text{Perf}_X}{\text{Perf}_Y} = \frac{ET_Y}{ET_X}$$

Calculating Execution Time

- Clock cycles and execution time



$$\text{Execution Time for a program} = \#\text{ of clock cycles} \times \underbrace{\text{clock cycle time}}_{\text{period}}$$

$$= \frac{\#\text{ of clock cycles}}{\text{clock rate (frequency)}}$$

Example 1

Machine X can perform a task in 11 seconds. Machine Y can perform the same task in 17 seconds. How much faster is X than Y?

$$N = \frac{ET_Y}{ET_X} = \frac{17}{11} = 1.54 \text{ times}$$

CPI = cycles per instruction

different insts. take a differing number of clock cycles { add, beq, lui, lw }

$$\text{total \# of CPU clock cycles} = \sum_{i=1}^{\text{all inst. classes}} (CPI_i \times C_i)$$

C_i: inst. count (# of insts. in a given class)

Example 2

	Instruction Counts		
Code sequence	A	B	C
old gcc 1	3	1	3
new gcc 2	0	3	3

Inst. class	CPI
arithmetic → A	1
branches → B	2
lw/sw → C	4

2 different compilers produce code sequences 1 and 2. Which is faster? What is the CPI for each sequence?

$$① \ \overbrace{1\cdot 3}^{A} + \overbrace{2\cdot 1}^{B} + \overbrace{4\cdot 3}^{C} = 17 \text{ clock cycles} \leftarrow \text{faster}$$

$$\text{Average } CPI_1 = \frac{\text{total cycles}}{\text{total inst}} = \frac{17 \text{ cycles}}{7 \text{ inst.}}$$

$$② \ \overbrace{1\cdot 0}^{A} + \overbrace{2\cdot 3}^{B} + \overbrace{4\cdot 3}^{C} = 18 \text{ clock cycles}$$

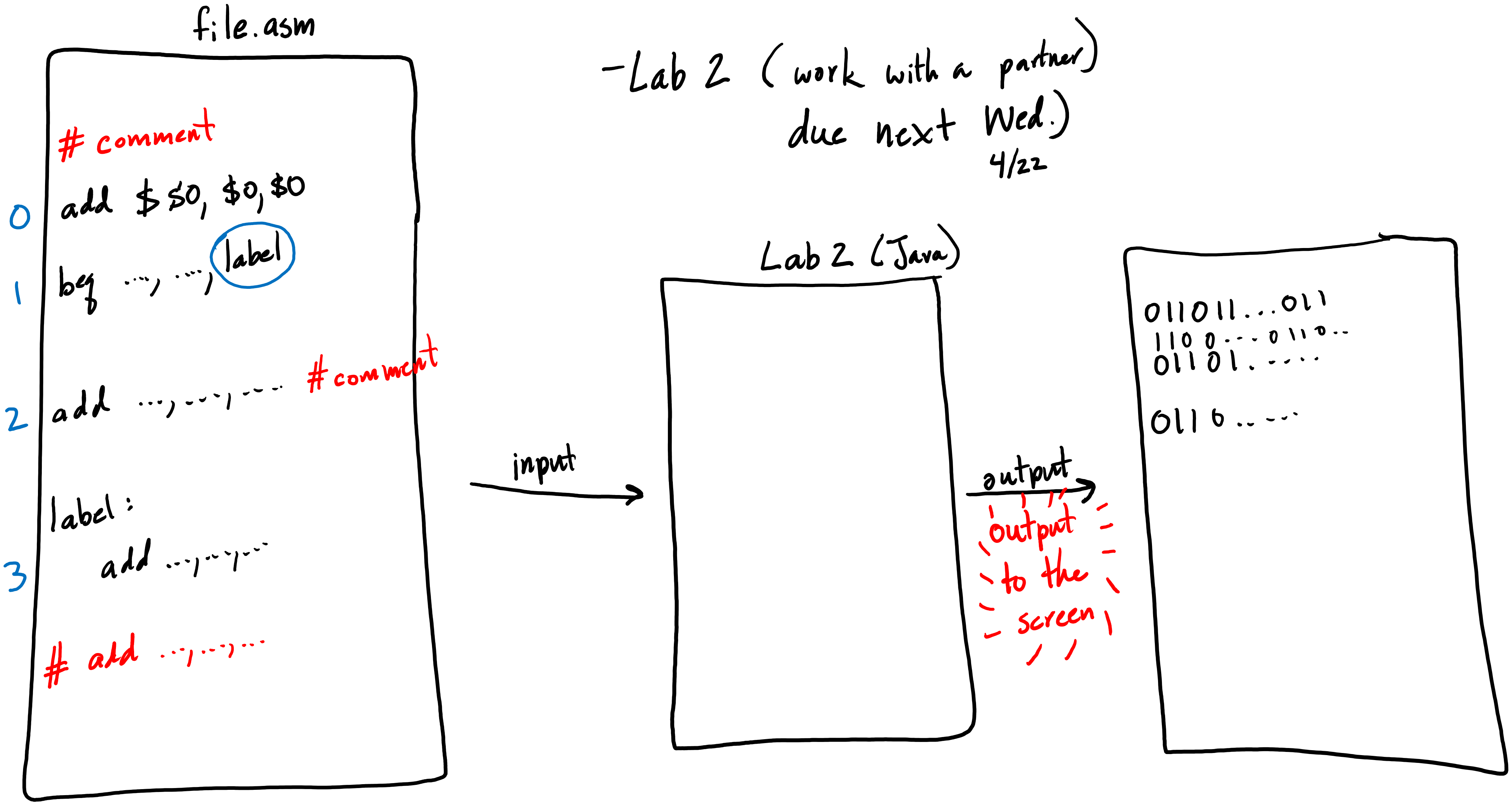
$$\text{Avg. } CPI_2 = \frac{18}{6} = 3$$

Calculating Execution Time

$$\text{Exec. Time} = \overbrace{\text{Inst. Count} \times \text{Avg. CPI}}^{\text{total \# of clock cycles}} \times \text{clock cycle time}$$

total # of inst. run

- Lab 2 (work with a partner)
due next Wed.)
4/22



Two pass assembler:

1. find all the labels

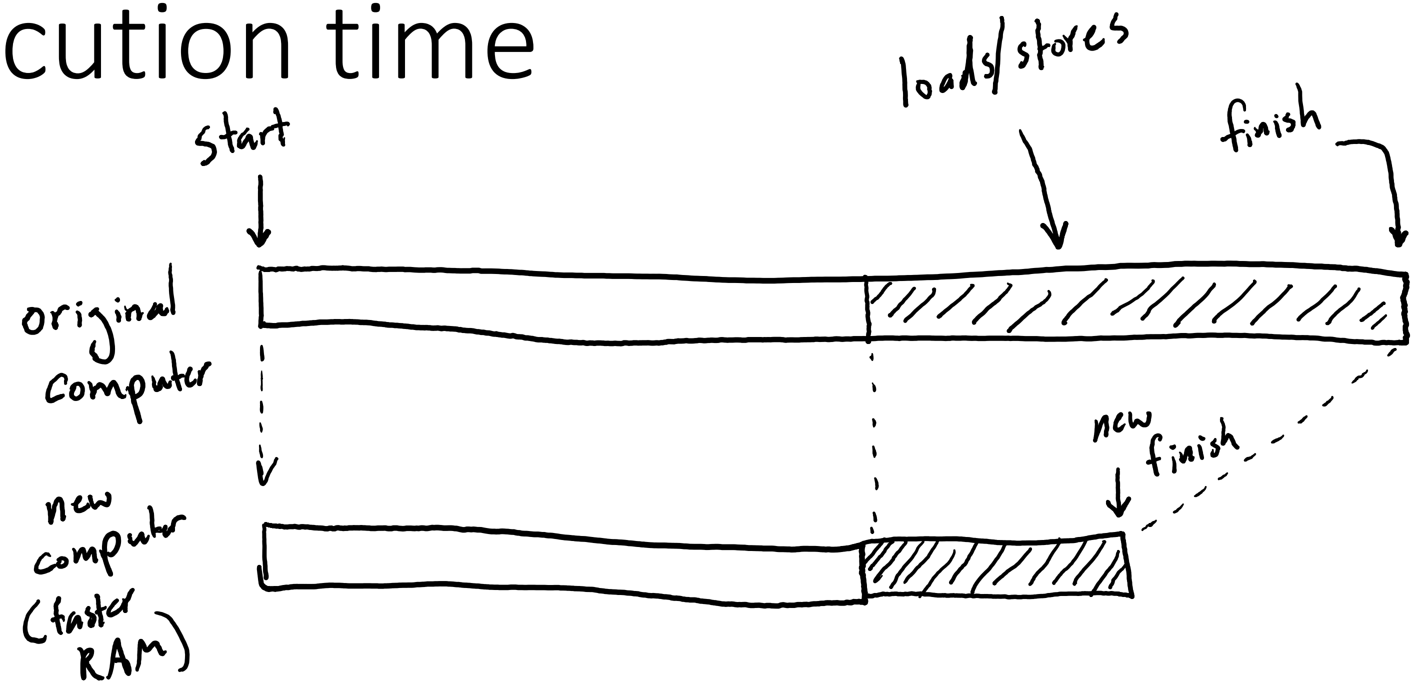
2. generate the binary output to the screen

diff $\underbrace{\text{-w -B}}_{\text{ignore whitespace differences}}$ your_output ref_output

- may be blank lines, whitespace
- all registers will be valid
- check for unsupported insts.
- handle decimal immediates

Improving execution time

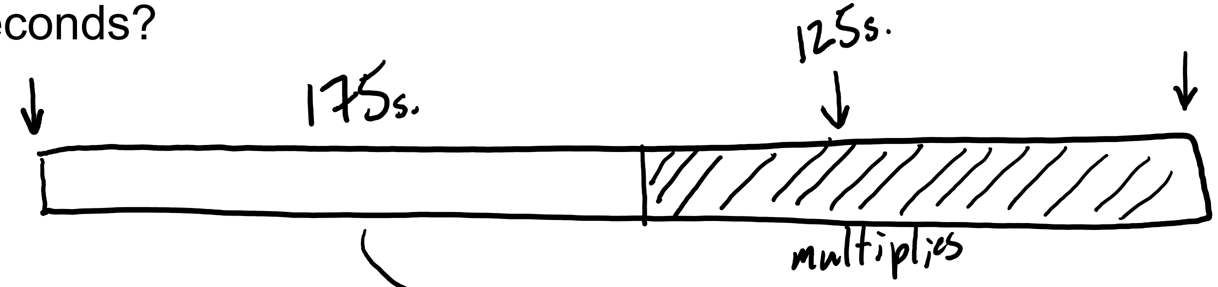
Amdahl's Law



$$\text{Exec. Time after improvement} = \frac{\text{Exec. Time affect by improvement}}{\text{Amount of Improvement}} + \text{Exec. Time not affected by improvement}$$

Example 3

A program takes 300 seconds to execute. Multiplies account for 125 seconds of the execution time. How much faster do multiplies have to be to execute program in 225 seconds?



$$225 = \frac{125}{N} + 175$$

$N = 2.5$ times